

Paul Dubois

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SUMMARY

“leveraging science for a better world”

Ph.D. in Applied Mathematics (STEM) applied to artificial intelligence for healthcare.

Expert in AI, data science, and teaching. Proficient in Python, Web (HTML/CSS, JavaScript), fast learner on new languages, with strong attention to detail and interpersonal skills.

EDUCATION

CentraleSupélec, Paris-Saclay, France

December 2024

Applied Mathematics, Ph.D. candidate

Concentration: Automation of treatment planning for radiotherapy using AI optimisation techniques.

Oxford University, Oxford, United Kingdom

July 2021

Master of Science, Mathematical Science

GPA: 68% (merit)

Key courses: Analytic Topology, Category Theory, Approximation of Functions, Theories of Deep Learning, Networks, Random Matrix Theory

Dissertation on Random Fractals and Branching Processes

University College London, London, United Kingdom

July 2020

Master of Science Integrated, Mathematical Science

GPA: 85% (first)

Key courses: Probability, Measure Theory, Spectral Theory, Functional Analysis, Multivariable Analysis, Differential Geometry, Analytic Number Theory, Graph Theory and Combinatorics, Elliptic Curves, Commutative Algebra, High-Performance Computing, Evolutionary Games and Population Genetics
Research project on Modular forms mod 2: “Governing Fields for the Hecke Algebra”

SKILLS & CERTIFICATIONS (Languages/Technology)

- Proficient in Python, Web Applications (HTML/CSS, vanilla JS, React, Vue), web scraping.
- Intermediate in SQL, C/C++, Android Development, IoT/electronics, CAD.

EXPERIENCE

PRF Development, Paris, France

October 2021 – Present

AI Engineer & Data Scientist

- *Client: Wheere (<https://www.wheere.com/>)*
Trained developed AI models for signal processing.
 - Improved accuracy of the system (25% gained accuracy within 5 months)
 - Created various reports for the clients.
- *Client: Surgitec (<https://www.surgitec.com/>)*
Trained models for the segmentation of sensitive regions in the brain using MRI (identified by surgeons as “applicable for clinical treatment”)
 - Developing AI software for Deep Brain Stimulation (DBS).
 - Designed a cost function for ranking candidate trajectories for DBS.
 - Created a new algorithm for best trajectory selection.
- *Web scraping for Acceor (<https://www.acceor.com/>) & Economics researcher from TSE.*

RoomFurnisher – London, UK

July 2020 – September 2020

Engineer (Intern)

- Developed the startup’s proof of concept by integrating open-source 3D reconstruction tools. Used **Python** and **AliceVision (Meshroom)** to perform **photogrammetry**, generating accurate 3D models of indoor environments from image datasets.

Airbus – Toulouse, France

June 2019 – September 2019

Data Scientist (Intern)

- Developed an internal **performance monitoring dashboard** from scratch using **SQL**, **HTML/CSS**, and **JavaScript** to visualize team productivity and key performance indicators (KPIs) metrics.
- Designed and deployed **Python-based automation tools** to streamline data collection and reporting workflows, reducing manual effort and improving data reliability across departments.

TEACHING EXPERIENCES

CentraleSupélec University, Paris, France

Engineering school

Lecturer

January - March {2023, 2024, 2025}

- Deep Learning & Natural Language Processing
Taught from scratch (back-propagation), perceptron, convolutions, optimizers & learning rates, RNN, U-net, V-net, GANs; with Kagglers created as homework.
Each class contains approximately 35 students (per semester).
Students' Feedback: "Ranked as No.1 among the 3rd-year courses" (2025)

Teaching Assistant

Coding	<i>Fall {2021, 2022, 2024}</i>
Algorithms	<i>Fall {2023, 2024}</i>
Reinforcement Learning	<i>Fall 2024</i>
Informatics	<i>Fall 2023</i>
NLP project	<i>Fall 2022</i>
Computer vision	<i>Spring 2022</i>
Optimization	<i>Fall 2021</i>

Essec University, Cergy, France

September {2021, 2023, 2024, 2025}

Business school

External lecturer

- Mathematics
Basic methodologies for proofs; linear algebra, differential calculus, integration, and asymptotic analysis; prerequisites of the courses in the master's.
Average satisfaction > 4,75/5 every year

Bootcamp Ironhack Barcelona, Spain

July 2023 - August 2023

Private school

Lead teacher/lecturer

- Data Science
Basic data science using Python, SQL, web technologies, Tableau and basic mathematics. *The content covers the premises of programming, data scraping, mathematical modelling, and artificial intelligence. The main resources are property of IronHack, but extra resources (mainly on AI, created by myself) are open source.*

PUBLICATIONS & POSTERS

Research Interest: AI, Front-end AI, Computer Vision, National Language Models

Google Scholar: [Paul Dubois - Google Scholar](#)

1. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos (2024). Radiotherapy Dose Optimization via Clinical Knowledge-based Reinforcement Learning. Artificial Intelligence in Medicine (AIME), (Salt Lake City, Utah, USA). https://dl.acm.org/doi/10.1007/978-3-031-66538-7_16
2. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos (2024). Clinically Dependent Fully Automatic Treatment Planning System. American Society for Radiation Oncology (ASTRO), (Washington, DC, USA). [https://www.redjournal.org/article/S0360-3016\(24\)02860-8/fulltext](https://www.redjournal.org/article/S0360-3016(24)02860-8/fulltext).
3. **P. Dubois**, Nikos Paragios, Paul-Henry Cournède, Gizem Temiz, Rafael Marini-Silva, Norbert Bus, Pascal Fenoglietto (2024). A Novel Framework for Multi-Objective Optimization and Robust Plan Selection Using Graph Theory. <https://user-swndwmf.cld.bz/ESTRO-2024-Abstract-Book/3687/> European Society for Radiotherapy and Oncology (ESTRO), Glasgow, UK.
4. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos. (2024). Dose Volume Histograms Guided Deep Dose Predictions. Société Française de Physique Médicale (SFPM), Dijon, France
5. **P. Dubois**, P. Fenoglietto, P.H. Cournède, N. Paragyos. (2024). Attention Mechanism on Dose-Volume Histograms for Deep-dose Predictions. Société Française de Radio-Oncologie (SFRO), La Défense, France.

AWARDS, HACKATHONS & COMPETITIONS

- **Science&Vie Junior Innovation Prize, February 2017** (1000€) - Morse translating machine.
- **Top 10 Prize, July 2017** - at the University College of London in mathematics.
- **Porticode 3.0 Fortnightly, November 2018** – Prize winner for best 90's-style website.
- **MakeHarvard 3D Modeling Helper, February 2019** – Prize winner; hardware translating real-world rotations to 3D software.
- **Ceteris Paribus Face Challenge, April 2022** – 3rd place; applied transfer learning to facial feature modification.
- **Hackathon ISEP Text-to-Speech, May 2022** – Winner; developed state-of-the-art text-to-speech AI.
- **Hackathon ISEP Green Code, October 2022** – Winner; optimized Python program for energy efficiency.
- **Hackathon ISEP FrHack, November 2022** – 3rd place; discovered statistically significant correlations between phone signals and radio waves.
- **Hackathon Essonne Eau & Numérique, April 2023** – 2nd place; simulated water levels in images using AI-generated depth maps.
- **Hackathon InterCentrales Leaf Nothing Behind, April 2023** – 4th place; predicted vegetation on occluded images with AI.
- **Citadel C1games Terminal Competition, May 2023** – 3rd place; coded AI strategy for competitive game tournament.
- **Hackathon ISEP IT4Green, January 2024** – Winner of 'Refurnished' prize; eco-friendly digital technology solutions.
- **Hackathon ISEP FrHack, March 2024** – 2nd prize; optimized frequency allocation for wireless cameras.
- **Hackathon Georgetown HoyaHack, January 2025** – Winner of Databricks prize; participated in Major League Hacking event.